Assessment 1 – Solutions

Course Code: CSA1704

Question 1: Water Jug Problem – Solution

To obtain exactly 2 gallons of water in the 4-gallon jug, a sequence of filling, pouring, and emptying operations is performed.

Steps:

Fill the 3-gallon jug completely.
Pour the water from the 3-gallon jug into the 4-gallon jug.
Fill the 3-gallon jug again.
Pour water from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug becomes full.
The 3-gallon jug now contains exactly 2 gallons of water.
Empty the 4-gallon jug.
Transfer the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug.

Final Result:
The 4-gallon jug contains exactly 2 gallons of water, which is the required goal state.

Question 2: Mars Rover Agent – Solution

The Mars Rover functions as an intelligent autonomous agent that senses its environment, processes information, and performs actions to achieve exploration objectives.

i) Percepts

The rover gathers information through:
Camera images
Soil composition data
Temperature readings
Atmospheric pressure measurements
Obstacle detection sensors
Wheel movement feedback

ii) Operating Environment

The rover operates in an environment that is:
Partially observable
Dynamic
Sequential
Stochastic
Real-world and uncertain

iii) Actions

The rover can perform actions such as:
Moving forward and backward
Turning left and right
Collecting soil and rock samples
Capturing images
Analyzing samples
Transmitting data to Earth

iv) Performance Measures

The rover's performance can be evaluated using:
Number of samples collected
Accuracy of scientific analysis
Distance explored
Energy efficiency
Mission success rate

v) Suitable Agent Architecture

Model-Based Goal-Oriented Agent

Reason:
Maintains knowledge about the environment.
Plans paths toward exploration targets.
Avoids obstacles effectively.
Makes decisions based on mission goals.
Adapts to changing environmental conditions.

Conclusion:
A Model-Based Goal-Oriented Agent is best suited for the Mars Rover because it combines environmental knowledge with goal-directed behavior to perform efficient exploration.

Question 3: 8-Queens Problem – Solution

The objective is to place eight queens on a chessboard so that no queen attacks another queen.

Problem Components

Initial State:
Empty 8 × 8 chessboard.

State Space:
All possible arrangements of queens on the chessboard.

Operators:
Place a queen in a safe position on the board.

Goal State:
A valid arrangement where:
No two queens are in the same row.
No two queens are in the same column.
No two queens are on the same diagonal.

Cost Function:
Each move has equal cost.

Search Strategy Used: Backtracking Search

Place a queen in the first column.
Move to the next column and place a queen in a safe position.
Continue this process for all columns.
If no safe position exists, backtrack to the previous column and try another position.
Repeat until all eight queens are safely placed.

Conclusion:
Backtracking efficiently explores the search space and finds a valid arrangement of eight queens without conflicts.

Question 4: OLA Cab Booking Application – Solution

The OLA Cab Booking System can be modeled as an intelligent agent that helps users reach their destination efficiently.

Type of Agent: Goal-Based Agent

Reason:

The primary goal is to transport the customer from source to destination.
The system evaluates available cab options.
It selects the most suitable cab.
It takes actions that help achieve the transportation objective.

Working Process:

Customer enters pickup location.
Customer enters destination location.
Available cab types are displayed.
Customer selects a preferred cab.
System checks driver availability.
If a cab is available:
Booking is confirmed.
Driver is assigned.
Trip begins.
Customer reaches destination.
Otherwise: Alternative cab options are suggested.

Conclusion:

The OLA booking application acts as a Goal-Based Agent because every decision is made to achieve the customer's goal of reaching the destination safely and efficiently.

Question 5: Uniform Cost Search (UCS) – Solution

Uniform Cost Search (UCS) is used to find the path with the minimum cumulative cost in a weighted graph.

Available Paths

Path	Cost
S → G	12
S → A → B → D → G	10
S → A → C → D → G	6
S → A → C → G	4

Working of UCS

Start at node S with cost 0.
Expand the node with the lowest path cost.
Add neighboring nodes to the priority queue.
Continue expanding the least-cost node.
Stop when the destination node G is reached.

Cost Calculation

S → G = 12
S → A → B → D → G = 1 + 3 + 3 + 3 = 10
S → A → C → D → G = 1 + 1 + 1 + 3 = 6
S → A → C → G = 1 + 1 + 2 = 4

Optimal Solution

Least-Cost Path:

S → A → C → G

Minimum Cost:
4

Conclusion:

Uniform Cost Search expands nodes in increasing order of path cost and guarantees the optimal solution. Therefore, the most economical route from warehouse S to warehouse G is S → A → C → G with a total cost of 4 units.

End of Assessment 1 Solutions